EXPERIMENT-1

LINUX PROGRAMMING ENVIRONMENT & BASIC COMMANDS

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1. Setting up of the platform

We can do C programming online or offline according to the needs. It is required for us to keep working on a Linux Environment.
Online: Cocalc.ai
Offline:
Methods: Dual boot, Single boot, Bootable Pendrive, VM

2. Virtual Machine Manager (Hypervisor)

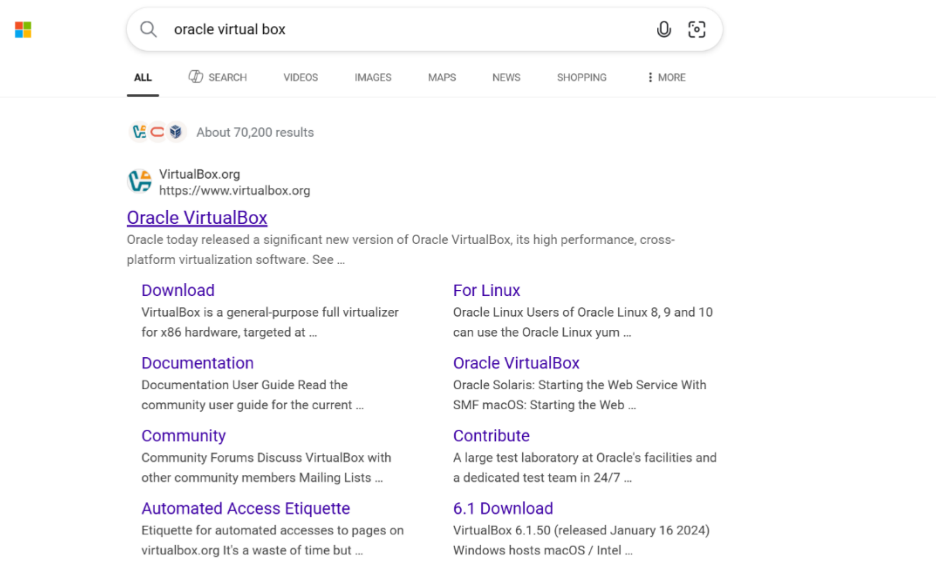
Type-1 Baremetal - **baremetal** means the **actual physical computer/server** that runs the virtual machines.
E.g. HyperV
Type-2 Hosted - It generally means that the virtualization software runs **on top of an existing operating system**, rather than directly on the physical hardware.

E.g Oracle Virtualbox, VM Ware.
We would be using Oracle Virtualbox.

3. Installation of Virtual Machine

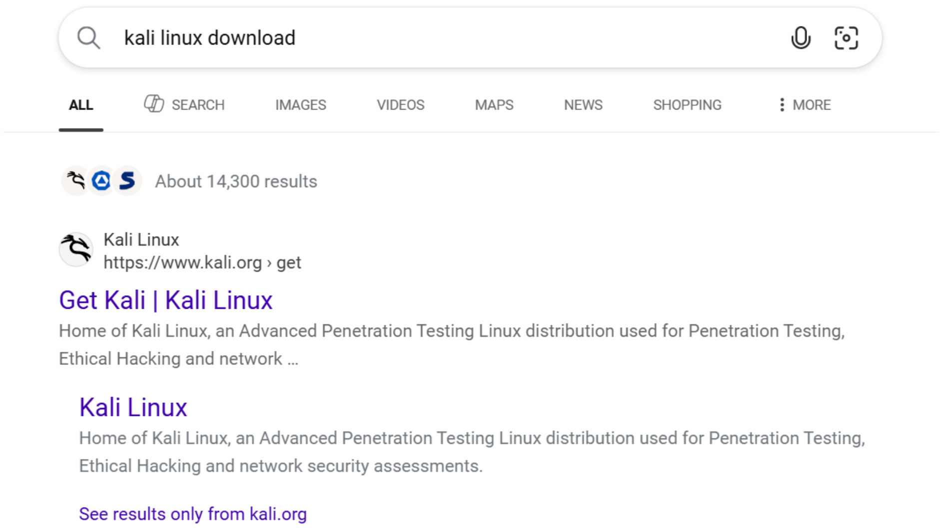
STEP-1: Download Oracle Virtual Box "windows hosts".

Link – https://www.virtualbox.org/



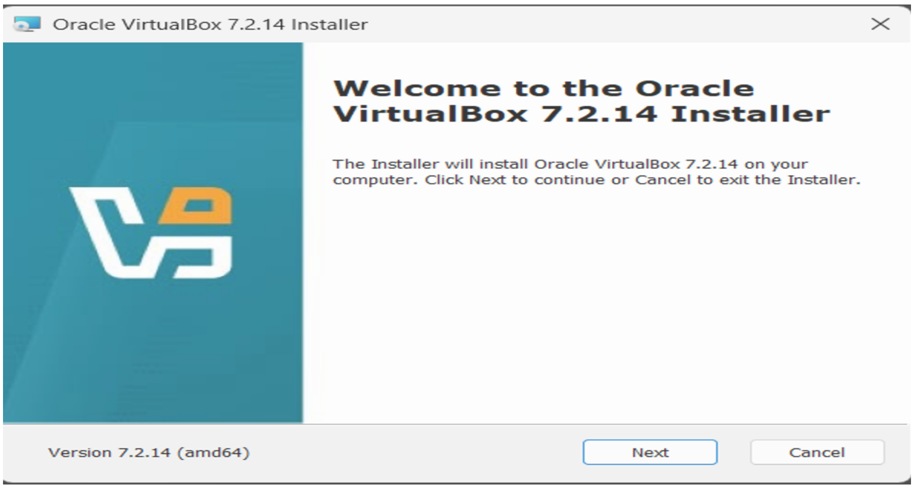
STEP- 2: Download Pre build Kali Linux Virtual Machine for Virtual box

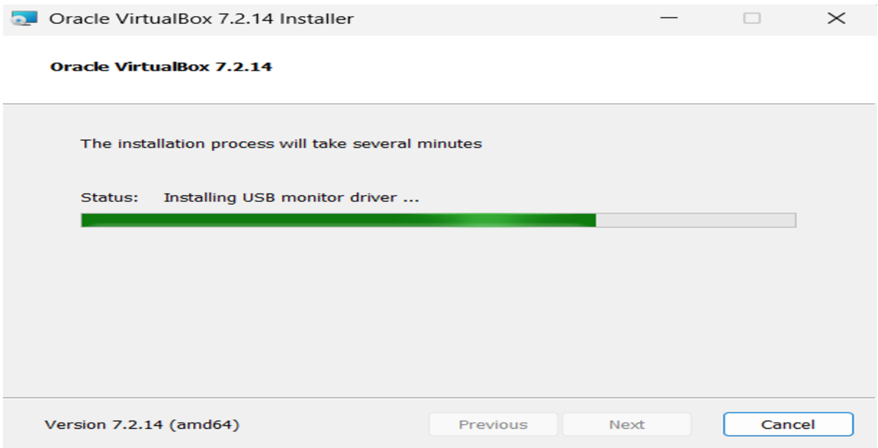
Link -https://www.kali.org/



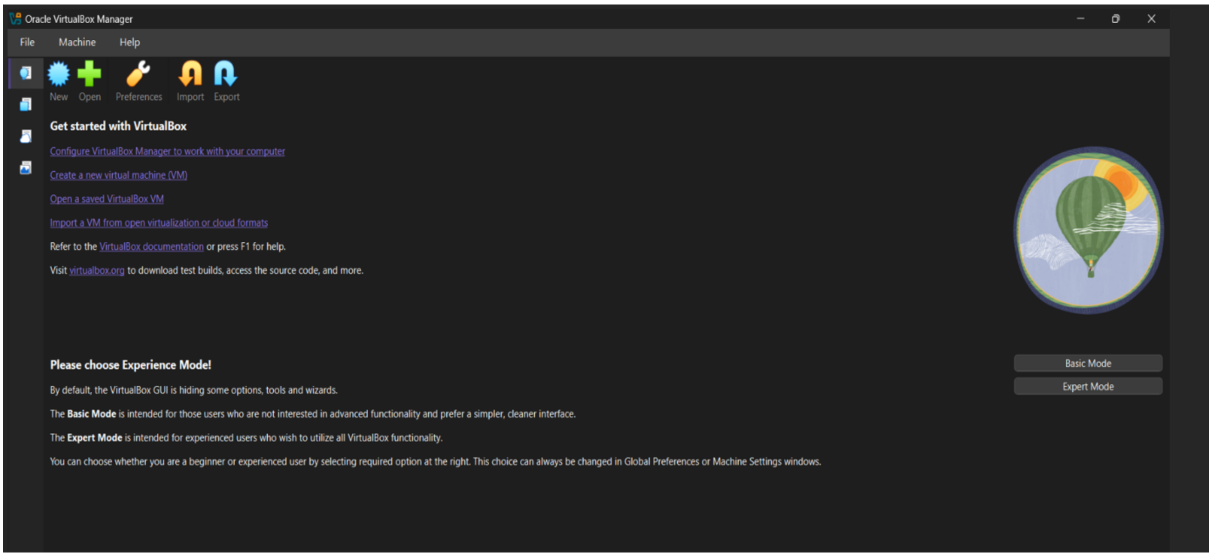
STEP-3: Extract the downloaded Kali Linux file

STEP-4 : Install the Virtual Box

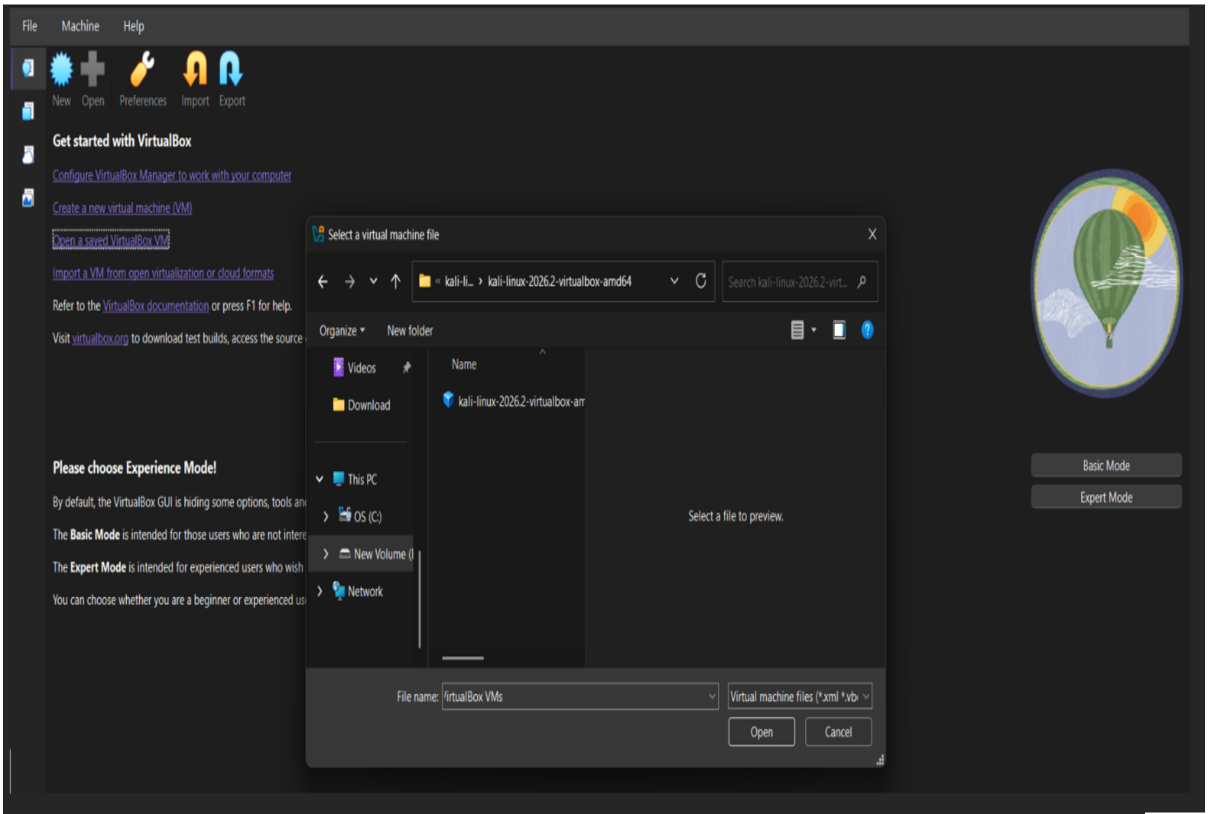


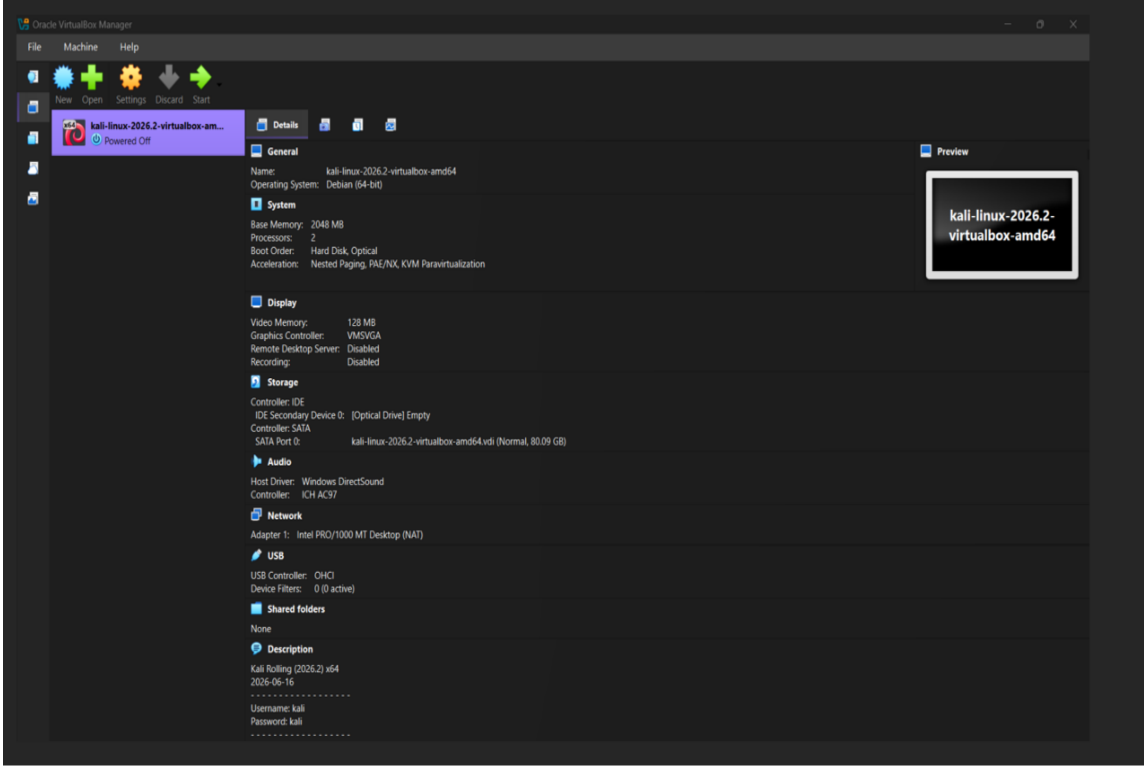


STEP-5: Open Virtual Box

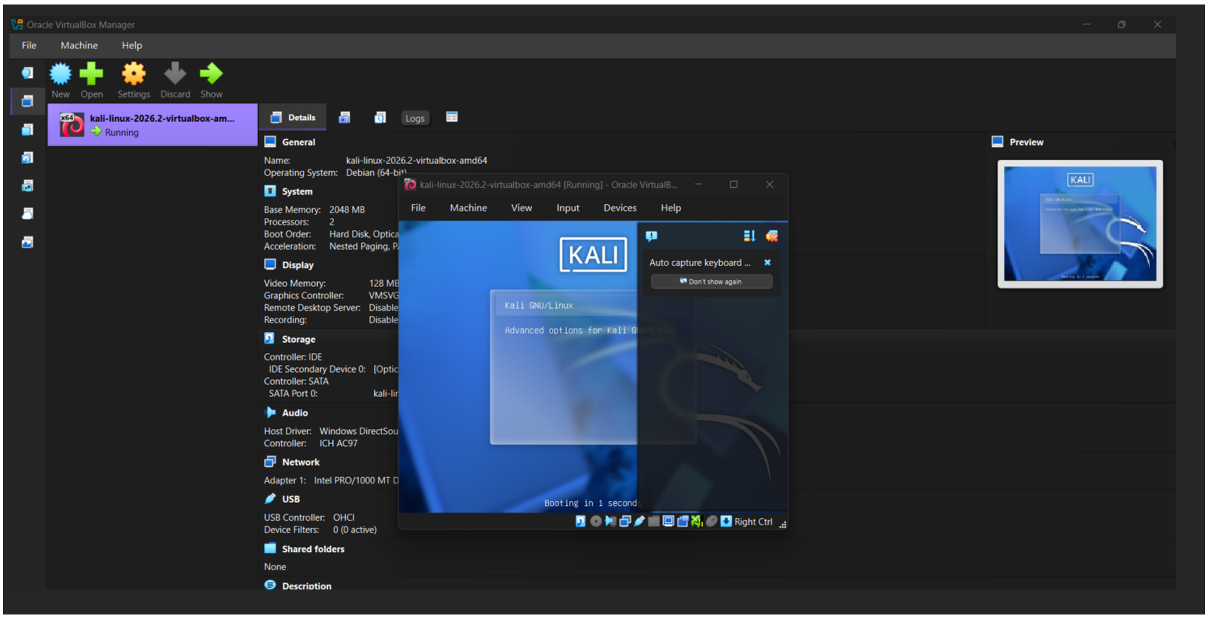


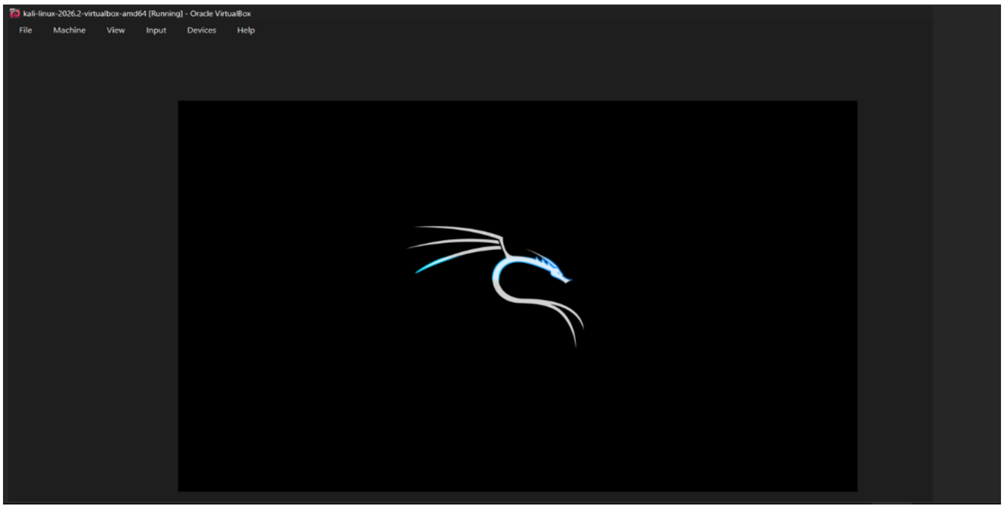
STEP-6: Opening the saved Virtual Machine

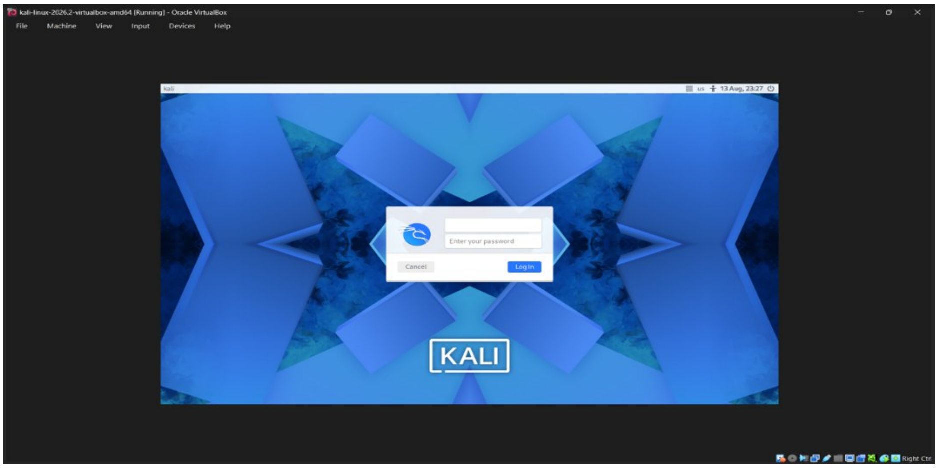




STEP-7: Start the Virtual Machine







Thus ,the installation of virtual machine manager(Hypervisor Type-2 Oracle Virtual Box) & Virtual Machine Kali Linux is completed.

STEP – 8 Openterminaland create the following simple program, compileand execute to understand the working of the environment.

Working with Linux Commands

Terminal basics:

COMMANDS	MEANING	EXAMPLE
mkdir	Create a new directory folder	mkdir notes.text
dir	List directoires	mkdir notes.text
ls	List directories	ls
rmdir	Remove an empty directory	rmdirtest
cd	Change directory	cd documents
cd..	Go up one directory	cd..
clear	Clear the terminal screen	clear
pwd	Print your current directory	pwd
History	Show all the previous used commands	history
man	Show the manualfor a command	man ls
whoami	Show the current logged in username	whoami

FILES COMMANDS:

COMMANDS	MEANING	EXAMPLE
vi	Create/edit a file	vi notes.text

file	Identify the type of a file	file notes.txt
Rm	Remove/delete a file	rm notes.txt
cp	Copy a file	cp a.txt b.txt
mv	Move or rename a file	mv a.txt b.txt
cat	Display file contents	cat notes.txt
more	View a file page by page	more notes.txt
less	View a file with the scrolling and search	less notes.txt
head	Show beginning of a file	head notes.txt
tail	Show end of a file	tail notes.txt
chmod	Change file permission	chmod 755 script.sh

REDIRECTION COMMANDS:

>	**Output redirection** – sends output to a file and **overwrites** existing content	echo "Hello" > file.txt
>>	**Append redirection** – sends output to a file and **adds** it to the end	echo "Kali" >> file.txt
<	**Input redirection** – takes input from a file and gives it to a command	sort < file.txt

All the above Commands working in kali linux:

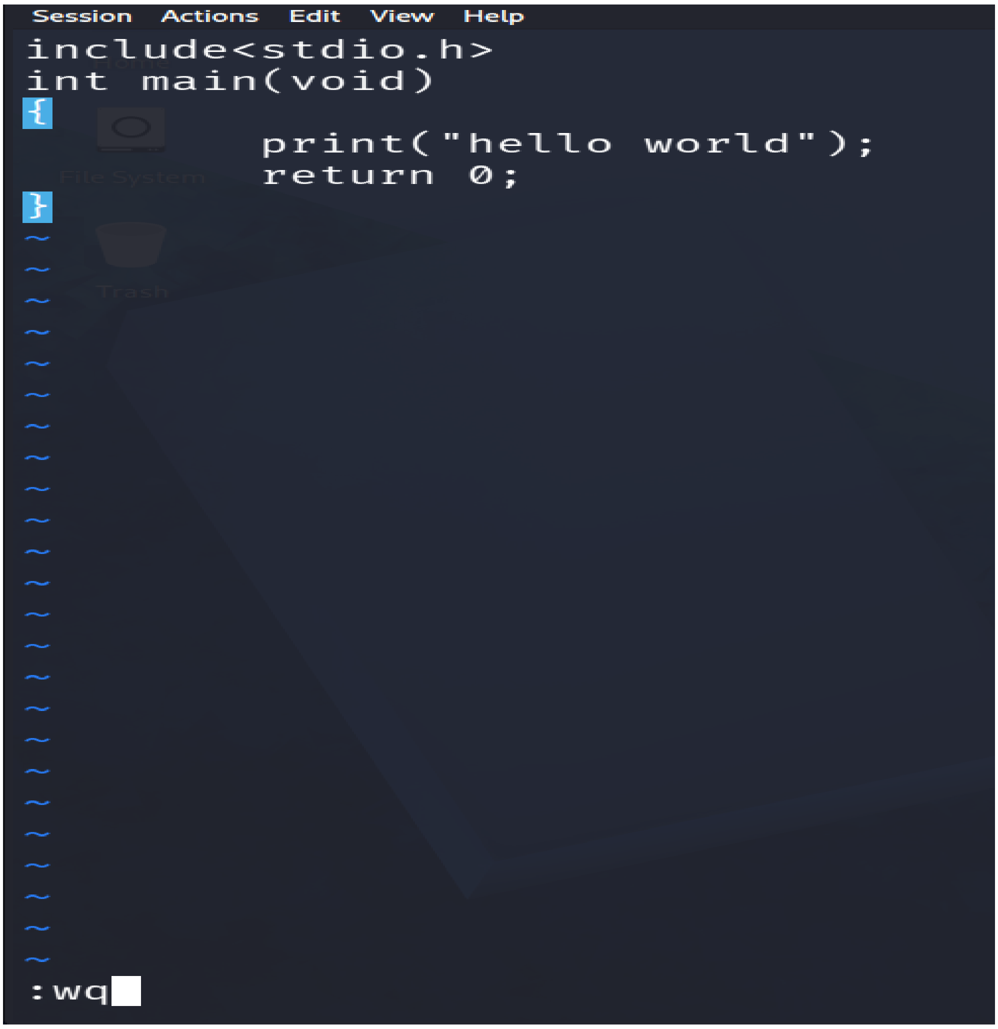
```
┌──(kali㉿kali)-[~]
└─$ pwd
/home/kali

┌──(kali㉿kali)-[~]
└─$ mkdir kali

┌──(kali㉿kali)-[~]
└─$ rmdir thru

┌──(kali㉿kali)-[~]
└─$ dir
area  Desktop  Documents  Downloads  kali  Music  muskan.c  notes.c  Pictures  Projects  Public  sunidhi  Templates  tripathi  Videos

┌──(kali㉿kali)-[~]
└─$ ls
area  Desktop  Documents  Downloads  kali  Music  muskan.c  notes.c  Pictures  Projects  Public  sunidhi  Templates  tripathi  Videos

┌──(kali㉿kali)-[~]
└─$ cd
```

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ whoami
kali

┌──(kali㉿kali)-[~]
└─$ man
What manual page do you want?
For example, try 'man man'.

┌──(kali㉿kali)-[~]
└─$ history
    1  pwd
    2  ls
    3  mkdie thru
    4  mkdir thru
    5  gedit area.c
    6  pwd
    7  ls
    8  mkdie thru
    9  mkdir thru
   10  gedit area.c
   11  pwd
   12  mkdir sunidhi
   13  ls
   14  #include<stdio.h>
   15  vi cnotes.c
   16  ped
   17  pwd
   18  mkdir
   19  mkdir sunidhi
   20  vi nidhi.c
   21  pwd
```

```
┌──(kali㉿kali)-[~/folder/folder]
└─$ touch file1.txt

┌──(kali㉿kali)-[~/folder/folder]
└─$ cp file1.txt copy.txt

┌──(kali㉿kali)-[~/folder/folder]
└─$ mv file1.txt new.txt
```

```
Session   Actions   Edit   View   Help
┌──(kali㉿kali)-[~]
└─$ mkdir folder

┌──(kali㉿kali)-[~]
└─$ cd folder

┌──(kali㉿kali)-[~/folder]
└─$ vi file1.txt
```

```
┌──(kali㉿kali)-[~/folder]
└─$ cd folder
cd: no such file or directory: folder

┌──(kali㉿kali)-[~/folder]
└─$ mkdir folder

┌──(kali㉿kali)-[~/folder]
└─$ cd folder

┌──(kali㉿kali)-[~/folder/folder]
└─$ touch file1.txt

┌──(kali㉿kali)-[~/folder/folder]
└─$ file file1.txt
file1.txt: empty

┌──(kali㉿kali)-[~/folder/folder]
└─$ rm file1.txt
```


Session Actions Edit View Help
include<stdio.h>
int main(void)
{
print("hello world");
return 0;
}
:wq


(kali㉿kali)-[~/folder/folder]
$ echo "hello">new.txt
(kali㉿kali)-[~/folder/folder]
$ cat new.txt
hello
(kali㉿kali)-[~/folder/folder]
$ echo "welcome">>new.txt